

# Tianyi Zhang

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## EDUCATION

**Georgia Institute of Technology**  
*Master of Science in Computer Science*

Atlanta, GA  
*Aug 2025 – Dec 2026*

**Emory University**  
*Bachelor of Science in Quantitative Sciences*

Atlanta, GA  
*Aug 2021 – May 2025*

## PROFESSIONAL EXPERIENCE

### LinkedIn Corporation

*AI/ML Engineer Intern | LLM for RecSys, LLM Post-Training*

Mountain View, CA  
*May 2026 – Present*

- Contributing to LinkedIn's **Apollo out-of-network retrieval** system for homepage-feed candidate generation, applying **LLM-based generative retrieval** to surface relevant content beyond a member's immediate network.
- Led a promising research direction on **adaptive and compressed member representation**, developing a framework for modeling profile signals, temporal engagement history, long-history compression, and member-regime differences in retrieval.
- Built a **multi-axis retrieval experimentation framework** to study tradeoffs among **Recall@K**, diversity, efficiency, retrieval data paradigm, and evaluation fidelity across cold-start, sparse-history, stale-history, and active-user scenarios.
- Developed a reusable **Apollo experimentation pipeline** for standardized training, retrieval evaluation, configuration reuse, execution tracking, and experiment logging, improving team velocity and reproducibility for LLM retrieval research.

### Georgia Tech Efficient & Intelligent Computing Lab · NVIDIA Collaboration

*Student Researcher | LLM Evaluation, Agentic AI*

Atlanta, GA  
*Jul 2025 – Dec 2025*

- Co-authored **COBA (ICML 2026)**, an automated pipeline auditing validity flaws in LLM-agent benchmarks; matched expert judgments at **0.79–0.87 F1** across **6 benchmarks** and showed flaws affect **~23% of tasks**, reshuffling **63% of model rankings** on a 30-LLM leaderboard.
- Built the **taxonomy-guided LLM-judge** stage and **rule-based structural detectors** powering the pipeline, decomposing agent tasks into user/environment/ground-truth/evaluation components; generalized to unseen benchmarks with no retuning and surfaced 100+ previously undocumented issues.

### Emory Graph Mining Lab

*Machine Learning Researcher | AI for Science, Graph Representation Learning*

Atlanta, GA  
*Dec 2023 – May 2025*

- First-authored a **causal brain-connectivity** method (**AIME 2025**) that recovered **directional** dependencies invisible to correlation-based graphs, improving downstream diagnostic prediction across **~8K patients** via Granger-directed graphs.
- Co-authored a **state-based Transformer (IEEE EMBC 2025, oral)** that modeled temporal transitions across latent functional-connectivity states, outperforming static-connectivity baselines on clinical prediction.

## PUBLICATIONS

H. Suh\*, S. Lee\*, B. Ji\*, R. Khare, B. Khan, H. Kim, **T. Zhang**, et al. “AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline” *International Conference on Machine Learning (ICML)* 2026.

**T. Zhang**, K. Han, J. Nie, et al. “Causal Brain Connectivity: Integrating Granger Directed Graphs in fMRI Analysis.” *International Conference on Artificial Intelligence in Medicine (AIME)* 2025.

J. Nie, K. Han, **T. Zhang**, et al. “Brain Network State Transformer: Leveraging State Functional Connectivity.” *IEEE Engineering in Medicine and Biology Society (EMBC)* 2025. (Oral)

## PROJECTS

### monitorstress: Stress-Test Runner for AI Safety Monitors 🐞

*Apr 2026 – Present*

- Architecting an **open-source evaluation framework** for **LLM-based AI safety monitors**—the deployed class of system labs like METR and UK AISI use to detect agent reward hacking and sandbagging—addressing a documented gap: monitors are evaluated mostly on clean transcripts, leaving robustness under structure-preserving transcript transformations under-measured.
- Designed and shipped the **Python/Pydantic v2** stack: typed event-stream trajectory model, verdicts with first-class abstention, adversarial transformation suite (chain-of-thought stripping, observation truncation, action-padding noise), **MALT** adapter (METR's public split of labeled agent transcripts), and a CLI runner reproducing METR's published monitor **prompt** as the reference baseline.

## TECHNICAL SKILLS

**Programming:** Python, SQL, Java, R, C/C++    **ML/AI:** PyTorch, CUDA, Hugging Face, vLLM, SGLang, LangChain, MLflow  
**Infra:** Docker, AWS, GitHub Actions, FastAPI, Linux    **Data:** PostgreSQL, MongoDB, Redis, Apache Spark